# Task 5: Exploratory Data Analysis (EDA)

## Titanic Dataset

### Introduction

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset. The analysis uses Python libraries such as Pandas, Matplotlib, and Seaborn to explore the dataset, identify patterns, trends, relationships, and potential anomalies.

The analysis includes statistical exploration, data visualization, missing value analysis, and correlation analysis to gain meaningful insights from the Titanic dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set the style for visualizations
sns.set_style("whitegrid")
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Cancel upload

```
df = pd.read_csv("train.csv")

df.head()
```

```
df.shape
```

```
df.tail()
```

```
df.info()
```

```
df.describe()
```

```
df.shape
```

```
df.isnull(). sum()
```

```
df.duplicated().sum()
```

```
df["Survived"].value_counts()
```

```
df["Cabin"].value_counts()
```

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Survived")

plt.title("Survival Distribution")
plt.xlabel("Survived")
plt.ylabel("Number of Passengers")

plt.show()
```

## Observation

The visualization shows the distribution of passengers who survived and those who did not survive. The number of passengers who did not survive was higher than the number of passengers who survived.

```
df["Sex"].value_counts()
```

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex")

plt.title("Gender Distribution of Passengers")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```

## Observation

The dataset contains more male passengers than female passengers, showing an unequal gender distribution among the recorded passengers.

```
df["Pclass"].value_counts()
```

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Pclass")

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```

## Observation

Passengers were distributed across three different ticket classes. The distribution helps identify how many passengers travelled in each class.

```
plt.figure(figsize=(8, 5))

sns.histplot(df["Age"].dropna(), bins=30, kde=True)

plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```

## Observation

The age distribution shows that passengers belonged to different age groups, with a large proportion of passengers being young adults. The distribution also helps identify the overall spread of passenger ages.

```
plt.figure(figsize=(8, 5))

sns.boxplot(x=df["Age"])

plt.title("Age Distribution and Outliers")

plt.show()
```

## Observation

The boxplot displays the median age and overall spread of passenger ages. It can also help identify potential outliers in the age variable.

```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival Based on Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```

## Observation

Female passengers had a higher survival rate compared to male passengers. This indicates that gender was an important factor associated with survival.

```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Pclass", hue="Survived")
```

```
plt.title("Survival Based on Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```

## Observation

Passenger class showed a noticeable relationship with survival. Passengers in higher classes generally had better survival outcomes compared with passengers in lower classes.

```
plt.figure(figsize=(8, 5))

sns.histplot(df["Fare"], bins=30, kde=True)

plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")

plt.show()
```

## Observation

The fare distribution is positively skewed, with most passengers paying lower or moderate fares and a smaller number of passengers paying significantly higher fares.

```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Survived", y="Fare")

plt.title("Fare Distribution Based on Survival")

plt.show()
```

## Observation

The fare distribution differs between passengers who survived and those who did not survive. This suggests that fare and passenger survival may have some relationship.

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(
    data=df,
    x="Age",
    y="Fare",
    hue="Survived"
)

plt.title("Age vs Fare Based on Survival")

plt.show()
```

## Observation

The scatterplot shows the relationship between passenger age and ticket fare while also displaying survival outcomes. The data points reveal variations in fares across different age groups.

```
sns.pairplot(
    df[["Survived", "Pclass", "Age", "Fare"]].dropna(),
    hue="Survived"
)

plt.show()
```

## Observation

The pairplot provides a comparison of multiple numerical variables and helps identify patterns, distributions, and possible relationships between them.

```
plt.figure(figsize=(8, 6))

numeric_df = df.select_dtypes(include=np.number)

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()
```

### Observation

The correlation heatmap shows the strength and direction of relationships between numerical variables. It helps identify positive and negative correlations within the dataset.

## Summary of Findings

The Exploratory Data Analysis of the Titanic dataset revealed several important patterns and relationships.

1. The dataset contains information about passengers, including age, gender, passenger class, fare, and survival status.
2. The number of passengers who did not survive was higher than the number who survived.
3. Female passengers generally had a higher survival rate than male passengers.
4. Passenger class showed a relationship with survival outcomes, with higher-class passengers generally having better survival chances.
5. The age distribution showed passengers from different age groups.
6. The fare distribution was positively skewed, with some passengers paying significantly higher fares.
7. Boxplots helped identify the distribution and potential outliers in numerical variables.
8. Scatterplots and pairplots helped explore relationships between numerical features.
9. The correlation heatmap provided insights into positive and negative relationships between numerical variables.
10. Overall, the analysis identified important patterns, trends, relationships, and potential anomalies in the Titanic dataset.

## Conclusion

Exploratory Data Analysis helped provide a better understanding of the Titanic dataset. Factors such as gender, passenger class, age, and fare revealed meaningful patterns related to passenger characteristics and survival outcomes. Visual and statistical exploration successfully identified trends and relationships within the data.